

FlexCube: 3D Printing Tunable Meta-structures with Triply Periodic Minimal Surfaces

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ABSTRACT

The development of additive manufacturing (AM) makes it possible to manufacture models with infill microstructures. Studies regarding microstructures that can alternating material properties have emerged. However, existing microstructure designs have many limitations for makers. Drawing inspiration from minimal surface structure, we demo a quickly designed and low-cost method to design and manufacture elastic structures with the flexibility of controlling mechanical properties in multiple directions. We also demo an interface to facilitate users to create FlexCube models. Based on previous researches and alternative perspectives, we explore a quickly designed and low-cost method to fabricate patterns with elastic properties, mainly based on the minimal surface structure.

CCS CONCEPTS

• **Human-centered computing** > **Human computer interaction (HCI)**;

KEYWORDS

3D printing, Microstructures, Deformable Objects, Interaction

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CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451562>

ACM Reference Format:

Lingyun Sun, Yu Chen, Deying Pan*, Yue Yang, Yitao Fan, Jiaji Li, Ziqian Shao, Ye Tao, and Guanyun Wang. 2021. FlexCube: 3D Printing Tunable Meta-structures with Triply Periodic Minimal Surfaces. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3411763.3451562>

1 INTRODUCTION

The development of additive manufacturing (AM) makes it possible to manufacture articles with internal microstructures. Studies that involve designing and redesigning microstructures on the subcentimeter scales have emerged; many are conducted to alternate the given physical properties of an object [6]. Additionally, concepts such as Metamaterial had been invented to implement structure design at a subcentimeter scale to achieve physical properties that are otherwise impossible [1, 2].

However, many existing microstructure design schemes mainly tackle 2-dimensional problems, while other 3-dimensional patterns suffer from limitations for their isotropicity, such as identical elasticities and Poisson's ratios under axial compression. Some studies even go as far as. Meanwhile, reproducing these procedurally generated structures requires knowledge of advanced algorithm design, modeling, and simulation. They are beyond the grasp of many novice makers, and manufacturing involves cost more than what is affordable for an average enthusiast [3–7].

In this demo, we propose a new method to design and manufacture elastic structures with the flexibility of controlling mechanical properties in multiple directions: starting with a predefined microstructure, we design custom features that resemble arrays of pillars in different directions and lengths. We also introduce a software interface to facilitate these elastic structures' design process and assess deformation's corresponding effect.

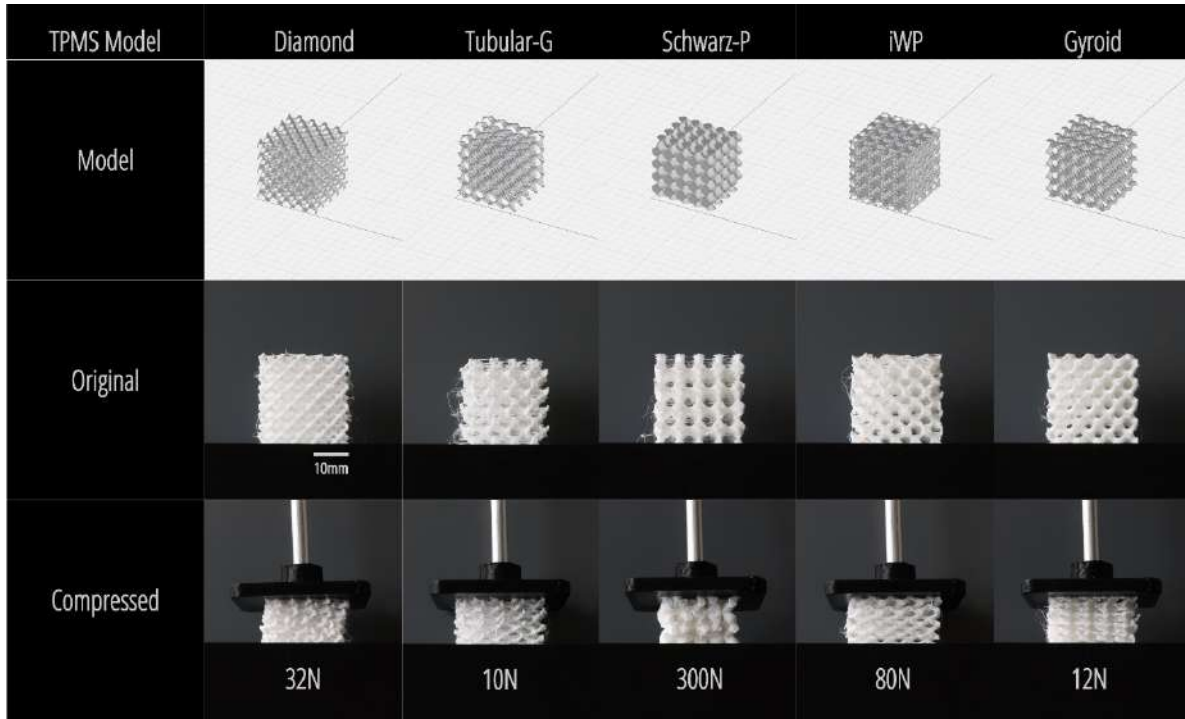


Figure 1: TPMS models and compression test.

We believe our work provides the necessary tools for modeling and manufacturing numerous types of flexible infill structures, and the methods we bring forth are efficient and cost-effective. This reflects our primary goal: to popularize personal fabrication in the name of the Do-it-yourself and the Maker Movement alike by utilizing the speed, flexibility, and consistency of 3D printing.

To design this structure-generating system, we took a look at the available schemes for our flexible base structures. The generation method we chose is Triply Periodic Minimal Surfaces (TPMS), a set of surfaces periodically arranged towards three orthogonal directions, and possess spatial symmetry in the Cartesian space [12]. In terms of 3D printing, TPMS, or minimal surface does not require support to fill up a volume with considerable height. Many equations can generate TPMS, so we pick five of the most representative designs, convert them into cubes, and produce them in 3D printers with thermoplastic polyurethane (TPU). Then, we conducted a series of compression tests to determine the forces required on each printed cube to have identical degrees of deformation. The result indicates that Tubular-G requires fewer materials and shorter printing times and performs well above average for its superior flexibility and softness (Figure 1).

2 SYSTEM DESCRIPTION

2.1 Primitive Library

We demonstrate how to design a system of actuators that changes our cube's physical properties. We used cylinders as the base primitives for simplicity, mechanical strength, and availability. Then we designed seven primitive units; each contains a unique pattern of

cylindrical pillars except for the first unit, which contains none. We then apply loads to each of the primitive units to find out the deformation degree and direction:

Primitive#2 has six pillars that are evenly spread out. As a result, the tensile strength in the compression direction is higher, and no non-uniform deformation is observed.

Primitive#3, primitive#4, and primitive#5 all have a set of six pillars that are tilted towards 30°, 45°, and 60°, respectively. Under 30N of loads, they display the inclined profiles of 20°, 26°, and 29°, respectively, which are consistent with our anticipation.

Primitive#6 has a similar design scheme, but unlike primitive#4, its pillars are oriented diagonally across the cube's volume, so the deformations are uniform towards both directions that are perpendicular to the load.

Primitive#7 and primitive#8 contain "C" shaped structures that are curved cylinders placed in a radial pattern, with the former convexed inward and the latter outward. Under the equivalent loads, #7 collapsed concentrically, while #8 collapsed eccentrically.

2.2 User Interface

We utilized Rhinoceros to prototype primitives, and we used Grasshopper as the foundation of our software. The interface contains four parts: Primitive Library, Settings, Simulation, and Export, in an order that matches the steps supposed to be taken by users (Figure 3). The user would first select the primitive by referring to the index numbers or the preview images; next, they can adjust by-unit settings by clicking the target unit and use the slider to perform axial translations and rotation; after the setup is finished, the user can preview the product by toggling the simulation switch

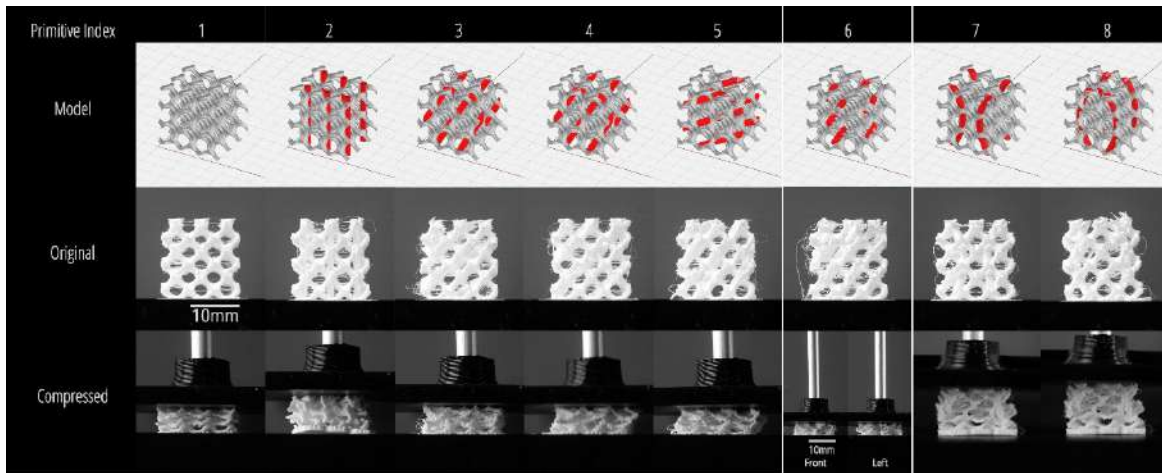


Figure 2: Primitive models and compression test.

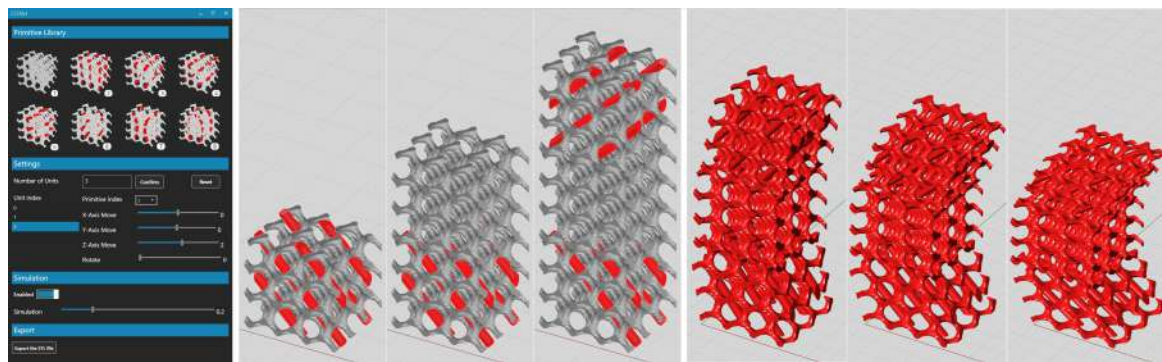


Figure 3: User interface (left), modeling process (mid), simulation (right)

to "Enabled," and then export the model in STL when a satisfactory result is met. In the example shown here, we use our software to construct a 1 unit by 1 unit by 3 units structure and have the simulation inform us about the predicted result once printed.

3 CONCLUSION AND FUTURE WORK

In the future, the software we envision can process the imported model from the user. It can automatically pixelate user-defined models and generate structures that match the predetermined deformation effect. As shown in the upper picture of figure 4, the original face is designed to deform into a smiley face, and the deformations are actuated in different directions. Included additionally in the bottom of the figure 4, the direction of columns in the middle branch is opposite to those in the branches of two sides, therefore the deformation direction is opposite as well. Furthermore, this system can even be used to fabricate user interfaces, and its controlled deformation can provide haptic feedback, which will be the key to bridge the gap between the physical and the virtual world. Besides, previous works have been done in interfaces that change shapes in different dimensions [8–11], and we suggested the possible method to control the deformation of the aforementioned work when exerting an external force.

ACKNOWLEDGMENTS

This project is supported by the National Science and Technology Innovation 2030 Major Project (2018AAA0100703) of the Ministry of Science and Technology of China, National Natural Science Foundation of China (No. 62002321), and the Provincial Key Research and Development Plan of Zhejiang Province (No. 2019C03137).

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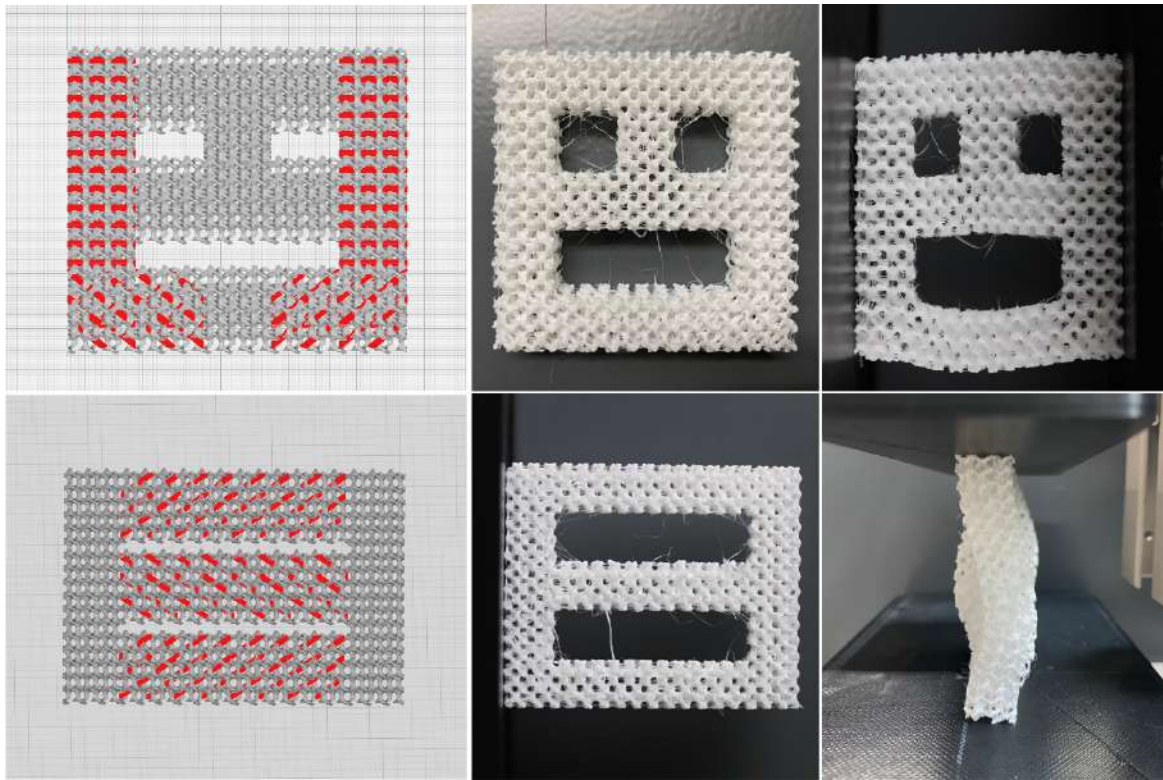


Figure 4: Applications

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